

Analysis of Variance (ANOVA)

STAT 24510-30040

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We have studied statistical tests to compare the means of two samples. Many experiments concern with the comparison of three or more samples or treatment groups. A common statistical method, the **analysis of variance** (ANOVA) is often the first step in comparing the means of multiple samples.

Data: The observations are g sets of measurements.

$$\begin{aligned} &\{y_{11}, \dots, y_{1n_1}\} \\ &\{y_{21}, \dots, \dots, y_{2n_2}\} \\ &\vdots \\ &\{y_{g1}, \dots, \dots, y_{gn_g}\} \end{aligned}$$

Question of interest: Did the g samples come from the same distribution, as opposed to, at least one sample having larger values than another sample on average?

Common data assumptions for simple ANOVA:

- Within each of the g samples, the data are independent observations from the same population.
- The g samples are independent of each other.

Under these assumptions,

- The range of index j in an observation y_{ij} depends on i .
- Observations in the same position in different samples may not carry useful meaning.
For example, y_{11} and y_{21} have no more in common than y_{11} and y_{23} .

Basics in Analysis of Variance

Historically, analysis of variance were applied to analyze treatment effects in comparative experiments.

A typical setting is a completely randomized experiment consisting of g treatments, or g treatment levels.

The g treatments have replications $r_1, \dots, r_g$, respectively.

$n = r_1 + \dots + r_g$ denotes the total number experimental units.

Notation and convention

- Random variables
We treat possible outcomes of the responses as random variables, commonly labeled with upper case letters.

Y_{ij} = the response random variable of the j th replication of treatment i .

$\bar{Y}_{i\bullet} = \bar{Y}_i = \sum_{j=1}^{r_i} Y_{ij}/r_i$ denotes the mean random variable of the i th treatment group.

$\bar{Y}_{\bullet\bullet} = \bar{Y} = \sum_{i=1}^g \sum_{j=1}^{r_i} Y_{ij}/n$ is the grand mean of all response variables.

- Observed data values

A common practice is to use lower case to denote the observed data, the realized values of random variables.

y_{ij} = the observed value of the j th replication of treatment i , a realization of random variable Y_{ij} .

$\bar{y}_{i\bullet} = \bar{y}_i = \sum_{j=1}^{r_i} y_{ij}/r_i$ denotes the observed mean value of the i th treatment group.

$\bar{y}_{\bullet\bullet} = \bar{y} = \sum_{i=1}^g \sum_{j=1}^{r_i} y_{ij}/n$ is the observed grand mean of all observations.

- In practice

It is common for statisticians to stop worrying about the distinction between the notations for random variables (generally upper case) and their realized values (generally lower case). The meaning of the notations should be based on the context.

- Other common notations:

J_i or n_i instead of r_i as group sample size; I instead of g as the number of treatment groups.

ANOVA objectives

Analysis of variance aims to address the question:

Do the treatments differ? That is, are there any treatment effects?

ANOVA methods answer the questions by analyzing the source of variations in the treatment outcomes.

One-way ANOVA or one-way layout is the case of one treatment factor with several levels.

For example, the treatments are different types of fertilizer, which can be treated as one treatment (fertilizer) with several levels (varieties of fertilizer). Another example is the comparison of difference vaccines for the same disease, where difference vaccines, including the placebo, are different levels of the vaccine treatment.

Two-way layout or high-way layout correspond to two or more treatment factors, each with several levels.

For example, in a two-way layout study, the two treatments can be various combinations of fertilizer and soil type, with several variety of fertilizer and several types of soil.

Sums of Squares in one-way ANOVA

In analysis of variance, the variations due to difference sources are compared.

In one-way analysis of variance, the total variations of the responses are partitioned into two parts:

the part due to the treatments, and the part due to random fluctuations.

The variations are measured by variance in terms of sums of squares.

- SS_{Trt} (a.k.a. $SS_{between}$, SS_B , SS_{model})

The variations of the treatment means (or group means) are described by the treatment sum of squares SS_{Trt} .

The **treatment** sum of squares (a.k.a. the **between** group sum of squares or the model sum of squares) is defined as

$$SS_{Trt} = \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^g r_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2$$

- SS_E (a.k.a. SS_{within} , SS_W , $SS_{residual}$)

Variations of individual observations from their own group mean are denoted by the error sum of squares SS_E .

The **error** sum of squares (a.k.a. the **within** group sum of squares or the **residual** sum of squares) is defined as

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

- SS_{total}

Variations of observations from the overall grand mean are described by the error sum of squares SS_{total} .

The **total** sum of squares is defined as

$$SS_{total} = \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$$

Variance-ratio and F -statistic for one-way ANOVA

Intuitively, if certain treatment or treatments have larger effects than other treatments, then the effects will make SS_{Trt} large, but should have no effects on SS_E .

The idea motivates the test statistic for ANOVA:

$$\text{Variance-Ratio} = \frac{SS_{Trt}/(g-1)}{SS_E/(n-g)} = F \quad (1)$$

The variance-ratio can be viewed as the ratio of 'signal' vs 'noise'.

The ratio should be large when some treatment means are significantly larger than others, and the variation is large when compared with the variation of random errors within treatment groups. On the other hand, the ratio should be relatively small when treatment mean variation does not stand out and is comparable to that of random errors.

We would like to evaluate the effects in probability statement, under certain assumptions.

Model assumptions and F-test

For the response of the j th replication of treatment i , Y_{ij} , a simple linear model for one-way ANOVA is

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2) \quad (2)$$

where

$$i = 1, \dots, g; \quad j = 1, \dots, r_i$$

Index i labels the treatment groups, r_i is the number of replications or the sample size of the i th group.

ε_{ij} are called residuals, errors or random deviations. A common pair of hypotheses of interest is

$$\begin{cases} H_o : & \mu_1 = \mu_2 = \dots = \mu_g = \mu \\ H_a : & \mu_i \neq \mu_j \quad \text{for some } i, j = 1, \dots, g. \end{cases}$$

Theorem

Under the assumption that the errors ε_{ij} are i.i.d. $\sim N(0, \sigma^2)$ in model (2), the null distribution of the variance-ratio statistic F in (1) is of F distribution with $(g-1, N-g)$ degrees of freedom.

Remarks on the ANOVA linear model

- The null distribution refers to the distribution under the null hypothesis H_o .
In one-way ANOVA, H_o represents the assumption that all treatment means are the same.
- Often there are observational studies for which the above model (2) might be used, even if the treatments are not randomly assigned.
- Recall that F distribution with degrees of freedom d_1 and d_2 is defined as $F = \frac{X/d_1}{Y/d_2} \sim F_{d_1, d_2}$, where $X \sim \chi_{d_1}^2$, $Y \sim \chi_{d_2}^2$, X and Y are independent.
 $\mathbb{E}(F_{d_1, d_2}) = \frac{d_2}{d_2-2}$ for $d_2 > 2$. Thus F_{d_1, d_2} takes values close to 1 when d_2 is large.
- The pair of hypotheses H_o, H_a also often denoted as H_0, H_1 .

Partition of variations in ANOVA

The sums of squares have the following relation.

$$\sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{\bullet\bullet})^2 = \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2 + \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})^2 \quad (3)$$

Proof.

$$\begin{aligned} \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{\bullet\bullet})^2 &= \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet} + \bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2 \\ &= \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2 + \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})^2 + 2 \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet}) \end{aligned}$$

Since for each $i = 1, \dots, g$,

$$\sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet}) = \sum_{j=1}^{r_i} Y_{ij} - r_i \bar{Y}_{i\bullet} = 0$$

Thus the cross product term sums to zero,

$$\sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet}) = \sum_{i=1}^g \left(\sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet}) \right) (\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet}) = 0$$

Hence (3) holds. $\square$

The partition of SS is commonly written as

$$SS_{Total} = SS_{Trt} + SS_E$$

with corresponding degrees freedom

$$\text{d.f. } n-1 = (g-1) + (n-g)$$

Interpretation

Total variations = variations due to treatments + variations due to random deviations or measurement errors

(or) = variations between treatments + variations within treatments

(or) = variations between groups + variations within groups

ANOVA table (One-way)

The results of the analysis of variance are commonly presented in an ANOVA table.

The following is an one-way ANOVA table, suitable for simple, unstructured, complete randomized design.

Source	SS (sum of squares)	d.f.	Mean SS	Variance ratio (F-value)
Treatments	$SS_{Trt} = \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2$	$g-1$	$SS_{Trt}/(g-1)$	$\frac{SS_{Trt}/(g-1)}{SS_E/(n-g)}$
Residuals	$SS_E = \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{i\bullet})^2$	$n-g$	$SS_E/(n-g)$	
Total	$SS_{Total} = \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$	$n-1$		

Remarks

- This is the format of ANOVA table used in common statistical software.
- Some computer software output leaves out certain redundancy in the table, such as (e.g., R) the last row, which can be derived from the rows above.
- Often there is another column $Pr(> F)$ at the end giving the p -value of the F -test for the variance ratio, under the null of equal treatment effects and normality assumptions.
- When the model assumptions do not hold such as in the case of many observational studies, the variance ratio is no longer of F distribution. Therefore the F variance ratio value and the p -value output should be for reference only.
- The **treatment mean square** (or mean of treatment sum of squares) is defined as

$$MS_{Trt} = SS_{Trt}/(g-1)$$

The **error mean square** is defined as

$$MS_E = SS_E/(n-g)$$

So the F -statistic can also be written as the ratio of mean squares.

$$F = \frac{SS_{Trt}/(g-1)}{SS_E/(n-g)} = \frac{MS_{Trt}}{MS_E}$$

- An alternative partition of variation is using non-centralized or crude sums of squares:

$$\sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij})^2 = \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{Y}_{i\bullet})^2 + \sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})^2, \quad \text{d.f. } n = g + (n-g).$$

This alternative partition is not as common as the mean-subtracted sums of squares that we used above.

Usually, the mean-subtracted sums of squares are used in the common ANOVA table.

Expected value of the sums of squares

Recall in one-sample case, if $X_1, \dots, X_n$ are *i.i.d.* with variance $V(X_i) = \sigma^2$, then the sample variance S^2 is an unbiased estimate of the true variance,

$$\mathbb{E}(S^2) = \mathbb{E}\left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right) = \sigma^2$$

Using the one-sample result, and the *i.i.d.* error assumption in the one-way ANOVA linear model (2), in which $Y_{ij} = \mu_i + \varepsilon_{ij}$, and ε_{ij} 's are i.i.d. $\sim N(0, \sigma^2)$, the expected value of the error sum of squares can be derived.

$$\begin{aligned} \mathbb{E}(SS_E) &= \mathbb{E}\left(\sum_{i=1}^g \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})^2\right) = \sum_{i=1}^g (r_i - 1) \mathbb{E}\left(\frac{1}{r_i - 1} \sum_{j=1}^{r_i} (Y_{ij} - \bar{Y}_{i\bullet})^2\right) \\ &= \sum_{i=1}^g (r_i - 1) \sigma^2 = \left(\sum_{i=1}^g r_i - g\right) \sigma^2 = (n - g) \sigma^2 \end{aligned}$$

Therefore the mean square for error is an unbiased estimator of σ^2 .

$$\mathbb{E}(MS_E) = \mathbb{E}\left(\frac{SS_E}{N - g}\right) = \sigma^2$$

Similar to the pooled sample variance for $g = 2$ samples, here MS_E is the pooled sample variance over all treatment groups.

$$MS_E = \frac{(r_1 - 1)S_1^2 + \dots + (r_g - 1)S_g^2}{n - g} = S_{pool}^2$$

To derive the expected value of treatment sum of squares, we define the true **grand mean** parameter μ as

$$\mu = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{r_i} \mu_i = \frac{1}{n} \sum_{i=1}^g r_i \mu_i.$$

Remarks on the grand mean vs average of group mean

- The **grand mean** is a weighted average of the group means.
- The **average of the group mean** is $\frac{1}{g} \sum_{i=1}^g \mu_i$.
- When the treatment sample size r_i 's are not equal, in general the grand mean is not the same as the average of the group mean.

$$\frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{r_i} \mu_i \neq \frac{1}{g} \sum_{i=1}^g \mu_i$$

- In the case of equal sample size $r_1 = \dots = r_g$, the two means are equal.
- When all g sample sizes are equal, other common notations are $r_1 = \dots = r_g = J$, $g = I$, so $n = IJ$.

Using the definition of the grand mean, we can derive the expected value for treatment sum of squares.

$$\mathbb{E}(SS_{Trt}) = \mathbb{E}\left(\sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2\right) = (g-1)\sigma^2 + \sum_{i=1}^g r_i (\mu_i - \mu)^2$$

Proof. By independence between groups,

$$Cov(\bar{Y}_{i\bullet}, \bar{Y}_{\bullet\bullet}) = Cov\left(\frac{\sum_j Y_{ij}}{r_i}, \frac{\sum_i \sum_j Y_{ij}}{n}\right) = Cov\left(\frac{\sum_j Y_{ij}}{r_i}, \frac{\sum_j Y_{ij}}{n}\right) = \frac{r_i}{n} Cov\left(\frac{\sum_j Y_{ij}}{r_i}, \frac{\sum_j Y_{ij}}{r_i}\right) = \frac{r_i}{n} V(\bar{Y}_{i\bullet}) = \frac{\sigma^2}{n}$$

Consequently,

$$\begin{aligned}\mathbb{E}(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2 &= V(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet}) + [\mathbb{E}(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})]^2 \\ &= V(\bar{Y}_{i\bullet}) + V(\bar{Y}_{\bullet\bullet}) - 2Cov(\bar{Y}_{i\bullet}, \bar{Y}_{\bullet\bullet}) + (\mathbb{E}\bar{Y}_{i\bullet} - \mathbb{E}\bar{Y}_{\bullet\bullet})^2 \\ &= \frac{\sigma^2}{r_i} + \frac{\sigma^2}{n} - 2\frac{\sigma^2}{n} + (\mu_i - \mu)^2 \\ &= \frac{\sigma^2}{r_i} - \frac{\sigma^2}{n} + (\mu_i - \mu)^2\end{aligned}$$

Hence

$$\mathbb{E}(SS_{trt}) = \sum_{i=1}^g \sum_{j=1}^{r_i} \mathbb{E}(\bar{Y}_{i\bullet} - \bar{Y}_{\bullet\bullet})^2 = (g-1)\sigma^2 + \sum_{i=1}^g r_i(\mu_i - \mu)^2$$

□

We then obtain the expected value of the treatment mean squares,

$$\mathbb{E}(MS_{Trt}) = \mathbb{E}\left(\frac{SS_{Trt}}{g-1}\right) = \sigma^2 + \frac{\sum_{i=1}^g r_i(\mu_i - \mu)^2}{g-1}$$

F test statistic for ANOVA

The above show that

$$\begin{aligned}\mathbb{E}(MS_E) &= \sigma^2 \\ \mathbb{E}(MS_{Trt}) &= \sigma^2 + \frac{\sum_{i=1}^g r_i(\mu_i - \mu)^2}{g-1}\end{aligned}$$

If we assume $\varepsilon_{ij} \sim N(0, \sigma^2)$, then SS_E/σ^2 is of χ^2 distribution with $n - g$ degrees of freedom:

$$SS_E/\sigma^2 \sim \chi_{n-g}^2$$

If in addition, $\mu_1 = \dots = \mu_g$ holds, SS_{Trt} is independent of SS_E , and

$$\mathbb{E}(MS_{Trt}) = \sigma^2, \quad SS_{Trt}/\sigma^2 \sim \chi_{g-1}^2 \quad \text{under } H_o$$

Therefore the variance ratio $\frac{MS_{Trt}}{MS_E} = \frac{SS_{Trt}/(g-1)}{SS_E/(n-g)}$ has null distribution of $F_{g-1, n-g}$.

The above provide us an appropriate test statistic for ANOVA hypothesis testing.

$$\begin{cases} H_o : & \mu_1 = \dots = \mu_g \\ H_a : & \mu_i \neq \mu_j \text{ for some } i, j = 1 \dots, g. \end{cases}$$

The F-test statistic

$$F = \frac{MS_{Trt}}{MS_E} \sim F(g-1, n-g) \quad \text{under } H_o.$$

Reject H_o when the test statistic is too large, as in the example of the resin data.

An example (ANOVA for resin glue failure times)

This example serves as a demo of hands-on analysis of variance.

- Experiment: The objective is to investigate the life time (in hours) of an encapsulating resin for gold-aluminum bonds in integrated circuits (operating at 120°C).
- Design: Conduct an accelerated life test. Randomly assign 37 units to one of five different temperature stresses.
- Variables

stress : levels of temperature stress (treated as categorical in this example)

Level 1	175°
Level 2	194°
Level 3	213°
Level 4	231°
Level 5	250°

logtime : $\log_{10}(\text{time})$, where time is the time-to-failure in hours of the tested material.

- Question: Does any treatment have an effect on the duration time?
- Data (partial)

	stress	logtime
1	1	2.04
2	1	1.91
3	1	2.00
4	1	1.92
5	1	1.85
6	1	1.96
7	1	1.88
8	1	1.90
9	2	1.66
10	2	1.71
11	2	1.42
12	2	1.76
13	2	1.66
14	2	1.61
15	2	1.55
16	2	1.66
17	3	1.53
...		
30	4	1.28
31	4	1.17
32	5	1.26
33	5	0.83
34	5	1.08
35	5	1.02
36	5	1.09
37	5	1.06

In this demo example, we treat the stress levels as categorical (even though there is an intrinsic order in the variable).

The overall mean of the logtime:

$$\bar{y}_{\bullet\bullet} = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{r_i} y_{ij} = \frac{1}{37} \sum_{i=1}^5 \sum_{j=1}^{r_i} y_{ij} = 1.465$$

The treatment means of logtime:

stress i	sample size r_i	logtime $\bar{y}_{i\bullet} = \frac{1}{r_i} \sum_{j=1}^{r_i} y_{ij}$
1	8	1.933
2	8	1.629
3	8	1.378
4	7	1.194
5	6	1.057

The **between** group sum of squares can be obtained,

$$\begin{aligned} SS_{Trt} &= \sum_{i=1}^g \sum_{j=1}^{r_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^5 r_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 \\ &= 8(1.933 - 1.465)^2 + 8(1.629 - 1.465)^2 + 8(1.378 - 1.465)^2 \\ &\quad + 7(1.194 - 1.465)^2 + 6(1.057 - 1.465)^2 \approx 3.54 \end{aligned}$$

To calculate the within group sum of squares:

$$\begin{aligned} \sum_{j=1}^{r_1} (y_{1j} - \bar{y}_{1\bullet})^2 &= (2.04 - 1.933)^2 + (1.91 - 1.933)^2 + \cdots + (1.90 - 1.933)^2 \\ \sum_{j=1}^{r_2} (y_{2j} - \bar{y}_{2\bullet})^2 &= (1.66 - 1.629)^2 + (1.71 - 1.629)^2 + \cdots + (1.66 - 1.629)^2 \\ &\vdots \\ \sum_{j=1}^{r_5} (y_{5j} - \bar{y}_{5\bullet})^2 &= (1.26 - 1.057)^2 + (0.83 - 1.057)^2 + \cdots + (1.06 - 1.057)^2 \end{aligned}$$

So the **within** group sum of squares can be obtained as the sum,

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{i\bullet})^2 = \sum_{i=1}^5 \sum_{j=1}^{r_i} (y_{ij} - \bar{y}_{i\bullet})^2 = \sum_{j=1}^{r_1} (y_{1j} - \bar{y}_{1\bullet})^2 + \sum_{j=1}^{r_2} (y_{2j} - \bar{y}_{2\bullet})^2 + \cdots + \sum_{j=1}^{r_5} (y_{5j} - \bar{y}_{5\bullet})^2 \approx 0.29$$

The observed F-statistic is

$$F = \frac{SS_{Trt}/(g-1)}{SS_E/(n-g)} = \frac{3.54/(5-1)}{0.29/(37-5)} \approx 97.66$$

The resulting F statistic is very large compared to 1. In fact, the p-value

$$P\{F(4, 32) \geq 97.66\} < 0.001$$

The data show strong evidence against the null hypothesis that all means are equal.

In fact, this simple demo example tells us only what we could tell from a data plot, that different treatments do have different means.

Sample R commands:

```
> stressfac=as.factor(stress)           # make stressfac as factors (categories), not numeric
> summary(aov(logtime~stressfac))
```

```
      Df Sum Sq Mean Sq F value    Pr(>F)    
stressfac  4  3.5376  0.88441   96.363 < 2.2e-16 ***
Residuals 32  0.2937  0.00918
```

Relation between ANOVA linear model and linear regression models

Recall the simple linear regression $Y_j = \alpha + \beta x_j + \varepsilon_j$, with independent observations $(x_j, y_j), j = 1, \dots, n$.

The estimated $\hat{\alpha}$ and $\hat{\beta}$, from a procedure such as Least Squares or Maximum Likelihood, give a "best fit line" $y = \hat{\alpha} + \hat{\beta}x$.

If there are g predictors $x_1, x_2, \dots, x_g$, the multiple regression model is

$$Y_k = \beta_0 + \beta_1 x_{1k} + \beta_2 x_{2k} + \cdots + \beta_g x_{gk} + \varepsilon_k, \quad k = 1, \dots, n. \quad (4)$$

To relate the common linear regression model (4) to the ANOVA linear model (2),

- Let β_i be the effect of treatment i for $i = 1, 2, \dots, g$.
- Let x_{ij} be the indicator function $1_{\{i\}}$ for the i th treatment,

$$x_{ij} = 1_{\{i\}} = \begin{cases} 1, & \text{if the } j\text{th experimental unit receives treatment } i, \\ 0, & \text{otherwise.} \end{cases}$$

- Relabel the responses within each group i as Y_{ij} (their observed values y_{ij}). Relabel the error terms similarly. Relabel β_i by μ_i . Set $\beta_0 = 0$,

$$Y_k = Y_{ij} = 0 + \cdots + \beta_i x_{ij} + 0 + \cdots + \varepsilon \implies Y_{ij} = \mu_i + \varepsilon_{ij}$$

Therefore the analysis of variance linear model (2) is a type of linear regression model.

Matrix representation, design matrix

The above argument can be more concisely represented using matrix notations.

$$Y_j = \alpha + \beta x_j + \varepsilon_j \quad j = 1, 2, \dots, n$$

is a shorthand for a set of n linear relationship or linear equations.

$$\begin{cases} Y_1 &= \alpha + \beta x_1 + \varepsilon_1 \\ Y_2 &= \alpha + \beta x_2 + \varepsilon_2 \\ &\vdots \\ Y_n &= \alpha + \beta x_n + \varepsilon_n \end{cases}$$

which can be written as

$$\begin{matrix} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} &= & \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} &+ & \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix} \\ \text{Dimensions:} & \begin{bmatrix} n \times 1 \end{bmatrix} & \begin{bmatrix} n \times 2 \end{bmatrix} & \begin{bmatrix} 2 \times 1 \end{bmatrix} & \begin{bmatrix} n \times 1 \end{bmatrix} \end{matrix}$$

By matrix multiplication,

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 \times \alpha + x_1 \times \beta \\ 1 \times \alpha + x_2 \times \beta \\ \vdots \\ 1 \times \alpha + x_n \times \beta \end{bmatrix} = \begin{bmatrix} \alpha + x_1 \beta \\ \alpha + x_2 \beta \\ \vdots \\ \alpha + x_n \beta \end{bmatrix}$$

For g predictor linear regression model

$$Y_j = \beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j} + \cdots + \beta_g x_{gj} + \varepsilon_j, \quad j = 1, 2, \dots, n$$

The matrix expression is

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{[n \times 1]} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1g} \\ 1 & x_{21} & x_{22} & \cdots & x_{2g} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{ng} \end{bmatrix}_{[n \times (g+1)]} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_g \end{bmatrix}_{[(g+1) \times 1]} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{[n \times 1]}$$

This is often written in more compact vector-matrix notation

$$Y = \mathbf{X}\beta + \varepsilon$$

where

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1g} \\ 1 & x_{21} & x_{22} & \cdots & x_{2g} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{ng} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_g \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The matrix $\mathbf{X}$ is called the **design matrix** of the model.

The Least Squares estimator for β is $\hat{\beta} = (X^T X)^{-1} X^T Y$, when $(X^T X)^{-1}$ exists, or when X is full-rank.

Example

An experiment has two treatment groups, each of 4 observations. Control responses $Y_{11}, Y_{12}, Y_{13}, Y_{14}$ have expected value (mean) μ_1 , similarly treatment responses $Y_{21}, Y_{22}, Y_{23}, Y_{24}$ have expected value μ_2 . A linear model in matrix form is

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{14} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ Y_{24} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{23} \\ \varepsilon_{24} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

Multiplying out the matrices gives

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, 2; \quad j = 1, 2, 3, 4$$

As mentioned before, commonly y_{ij} denotes observed realizations (data) of random variables Y_{ij} .

Example

An experiment has 3 treatment groups, each of 2 observations. the linear model is

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, 2, 3; \quad j = 1, 2,$$

which represents six linear relations.

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{31} \\ Y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

For experiments of more than two treatment groups, analysis of variance (ANOVA) is often the first step in data analysis.

MLE and ANOVA

The ANOVA linear model (2) $Y_{ij} = \mu_i + \varepsilon_{ij}$ indicates that maximum likelihood can be used to estimate the parameters. For each observation y_{ij} , its probability density function is a function of the unknown parameters.

$$f(y_{ij} | \mu_i, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_{ij} - \mu_i)^2}{2\sigma^2}}$$

The ANOVA linear model (2) with i.i.d. observations $y_{ij}, j = 1, \dots, r_i, i = 1, \dots, g$ has the model likelihood as a function of the parameters μ_i 's and σ^2 .

$$\ell(\mu_1, \dots, \mu_g, \sigma^2 | y_{ij}) = \prod_{i=1}^g \prod_{j=1}^{r_i} f(y_{ij} | \mu_i, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{\sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \mu_i)^2}{2\sigma^2}}$$

Maximum likelihood estimators can be derived as usual. As expected, the MLEs are (exercise)

$$\hat{\mu}_i = \bar{y}_{i\bullet}, \quad i = 1, \dots, g; \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{r_i} (y_{ij} - \hat{\mu}_i)^2 \quad (5)$$

Likelihood ratio test can be conducted to test the ANOVA hypotheses H_o of equal means and its alternative H_a .

Not surprisingly, the derivation should show that Likelihood ratio test is equivalent to the F test (exercise).

Alternative parameterization

The ANOVA linear model (2)

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \dots, g; \quad j = 1, \dots, r_i$$

often re-parameterized as an additive model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, g; \quad j = 1, \dots, r_i. \quad (6)$$

where α_i is called the "treatment effect" of the i th treatment. The likelihood function is re-parameterized as well.

$$\ell(\mu, \alpha_1, \dots, \alpha_g, \sigma^2 | y_{ij}) = \prod_{i=1}^g \prod_{j=1}^{r_i} f(y_{ij} | \mu, \alpha_i, \sigma^2)$$

where

$$f(y_{ij} | \mu, \alpha_i, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_{ij} - \mu - \alpha_i)^2}{2\sigma^2}}$$

Parameter redundancy

There is a redundancy in the number of the parameters in model (6). In other words, the model is over-parameterized. One parameter can be derived once other parameter values are fixed.

For example, when $g = 3, r_i \equiv 2$, the linear model

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \dots, g; \quad j = 1, \dots, r_i,$$

represents six linear relations.

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{31} \\ Y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

which is an expanded form of the linear regression model in matrix form

$$Y = \mathbf{X}\beta + \varepsilon$$

The design matrix $\mathbf{X}$ is of full rank 3. The unknown parameters are μ_1, μ_2, μ_3 and σ^2 .

On the other hand, the model (6)

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, g; \quad j = 1, \dots, r_i,$$

is an equivalent model, only re-parameterized, also represents six linear relations.

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{31} \\ Y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

In addition to σ^2 , now there are four unknown parameters $\mu, \alpha_1, \alpha_2, \alpha_3$. Since

$$\mu_i = \mu + \alpha_i$$

when any three of the four parameters $\mu, \alpha_1, \alpha_2, \alpha_3$ is known, the fourth can be derived.

The model (6) is over-parameterized. The redundancy also shows up in the not-full-rank of the design matrix (exercise).

Parameter constraints need to be imposed to make the parameters identifiable.

Common parameter constraints for model (6)

- Sum constraint $\sum_i \alpha_i = 0$, as in some textbooks.
- Baseline constraint on the first coefficient $\alpha_1 = 0$ as in the statistics software *R*.
- Baseline constraint on the last coefficient $\alpha_g = 0$, as in some other software.

In the above example of $g = 3, r_i \equiv 2$, the sum constraint

$$\alpha_1 + \alpha_2 + \alpha_3 = 0$$

can be expressed as

$$\alpha_3 = -\alpha_1 - \alpha_2$$

Now the unknown parameters to be estimated are μ, α_1, α_2 and σ^2 , the same number of unknown parameters as in the mean model (2). The re-parameterized model (6) under the sum constraint can be expressed as

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{31} \\ Y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

Note that now the design matrix $\mathbf{X}$ is of full rank.

In the same example of $g = 3, r_i \equiv 2$, the baseline constraint

$$\alpha_1 = 0$$

would result in

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \\ Y_{31} \\ Y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{bmatrix}, \quad \varepsilon_{ij} \sim i.i.d. \ N(0, \sigma^2)$$

Again the design matrix $\mathbf{X}$ is of full rank.

Estimates under various parameterizations

Parameter estimates obviously should vary with the way of parameterization.

- MLEs derived in (5) only apply to model (2), where $Y_{ij} = \mu_i + \varepsilon_{ij}$.
- In the reparameterized model (6), the MLE estimates for α_i depend on the type of constraints.
- All estimates should be equivalent. For example,

$$\mu_i = \mu + \alpha_i$$

implies

$$\hat{\mu}_i = \hat{\mu} + \hat{\alpha}_i$$

Analogously,

$$\mu_i - \mu_j = \alpha_i - \alpha_j$$

yields

$$\hat{\mu}_i - \hat{\mu}_j = \hat{\alpha}_i - \hat{\alpha}_j$$

- Estimated relative treatment effects and their significance should not change.

References: Sections 12.1, 12.2 in Rice, Chapter 11 in Casella and Berger chapter.